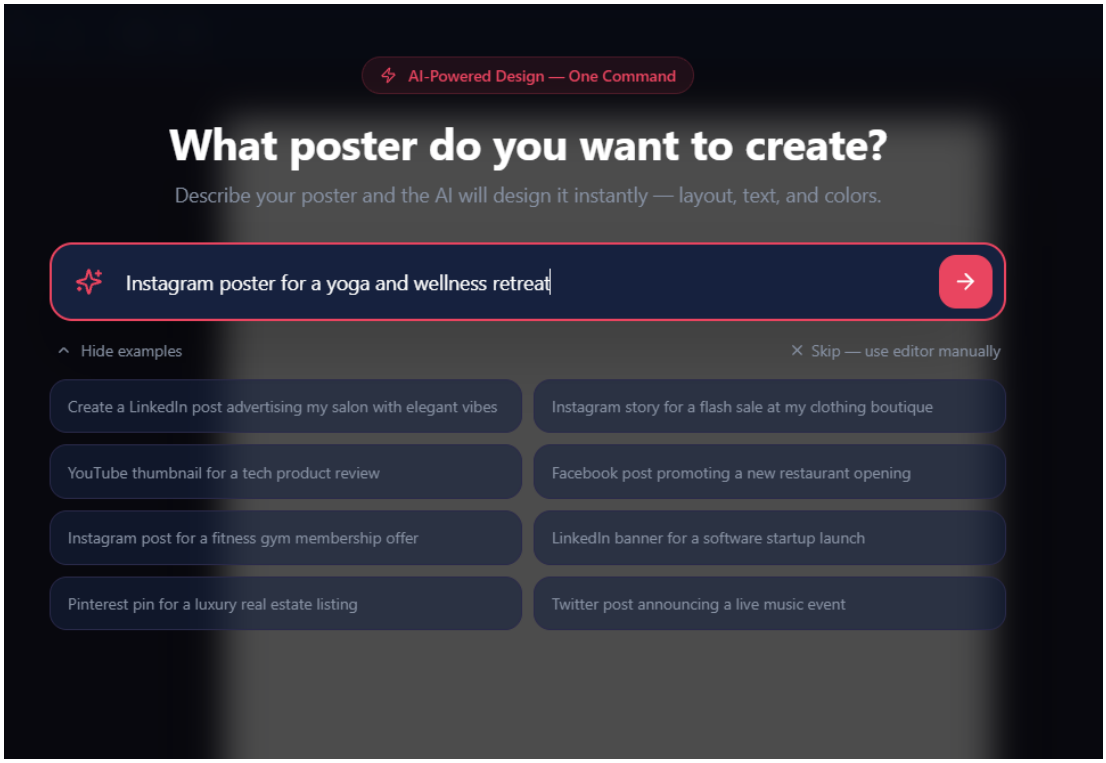
PosterPilot

PosterPilot is an agentic AI poster creation app where users type a single natural-language prompt (e.g., "create an Instagram post for my coffee cafe") and receive a complete, publication-ready poster with contextually relevant imagery, industry-tailored copy, and smart anti-overlap layout.
After generation, every element can be refined interactively on a drag-and-drop canvas.

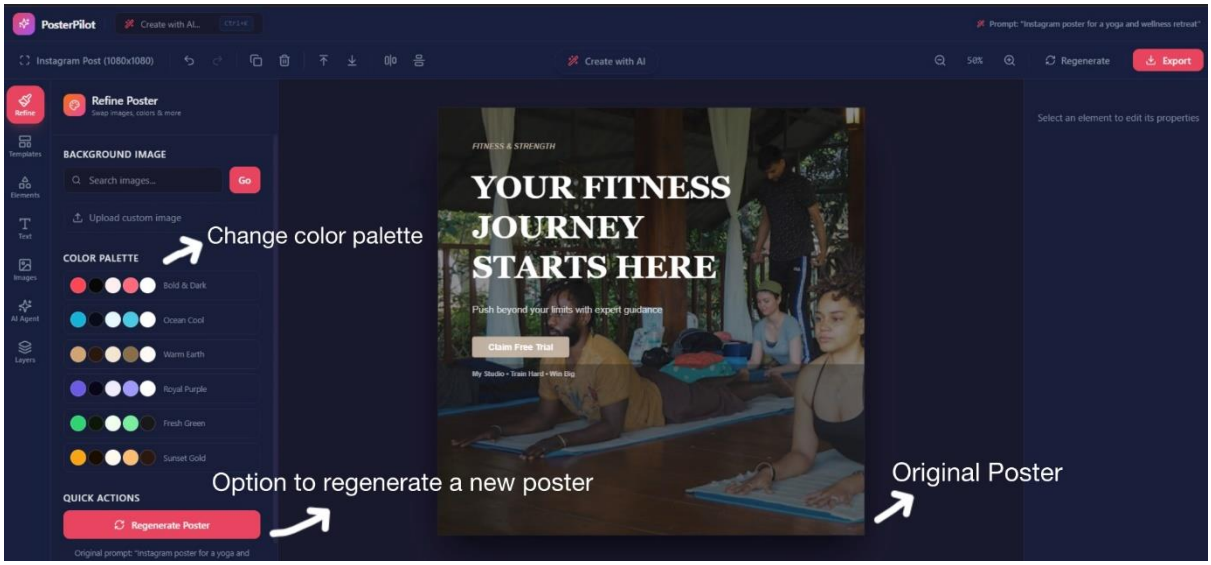
Key technologies:

Technology	Role
Next.js 14 (App Router)	Full-stack React framework with API routes
TypeScript	End-to-end type safety
Fabric.js 5	Canvas rendering, element manipulation, zoom, export
OpenAI GPT-4o-mini	AI content generation (headlines, copy, color palettes)
Unsplash + Pexels APIs	Dynamic, prompt-relevant background image sourcing
Zustand	Lightweight global state management
Tailwind CSS	Utility-first styling
Vercel AI SDK	Streaming AI responses

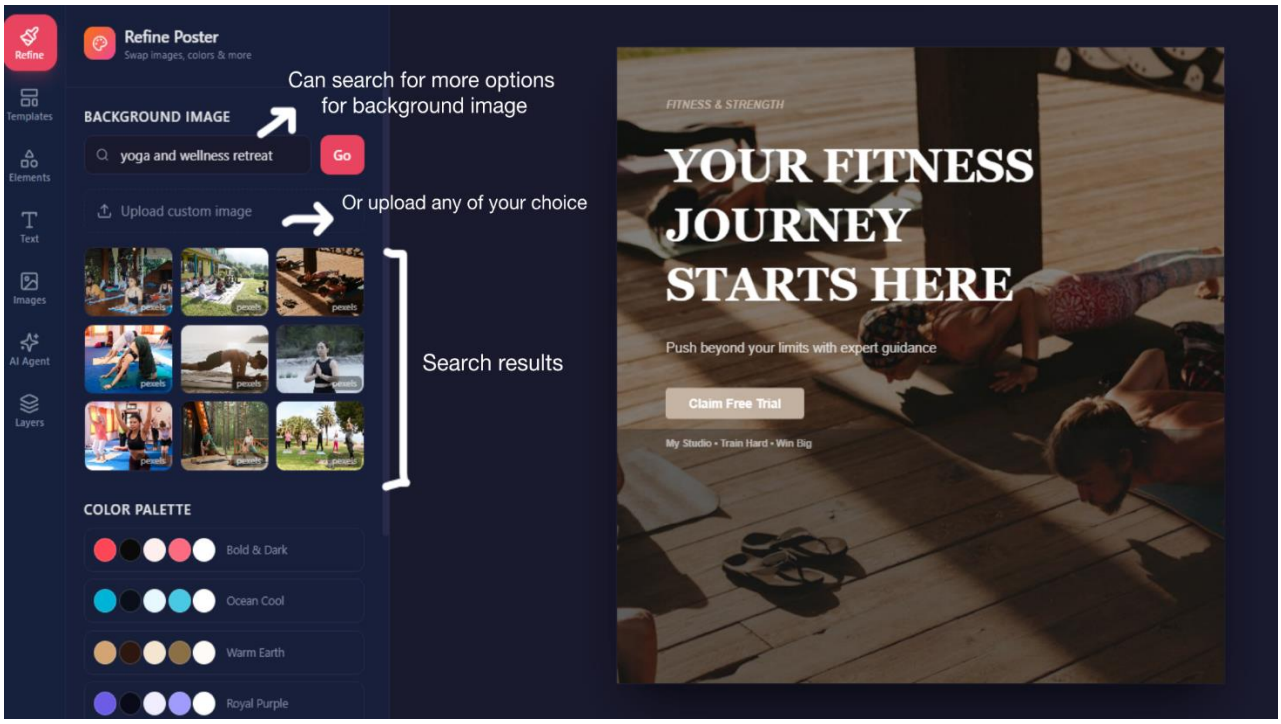
This is what Search Home Page looks like, with added examples, and the option to manually create poster-



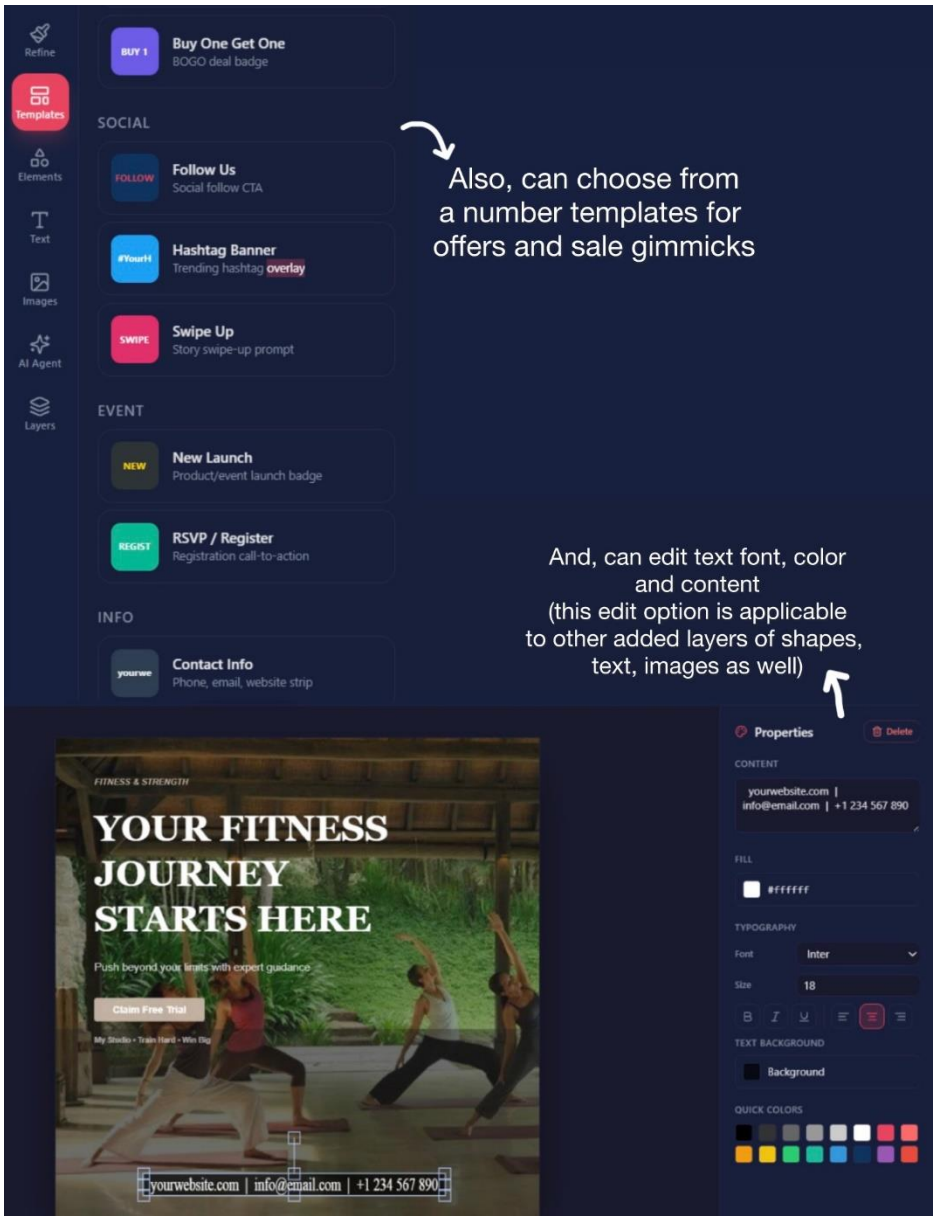
The Agentic AI digs for the perfect poster and comes up with a screen like below in seconds-



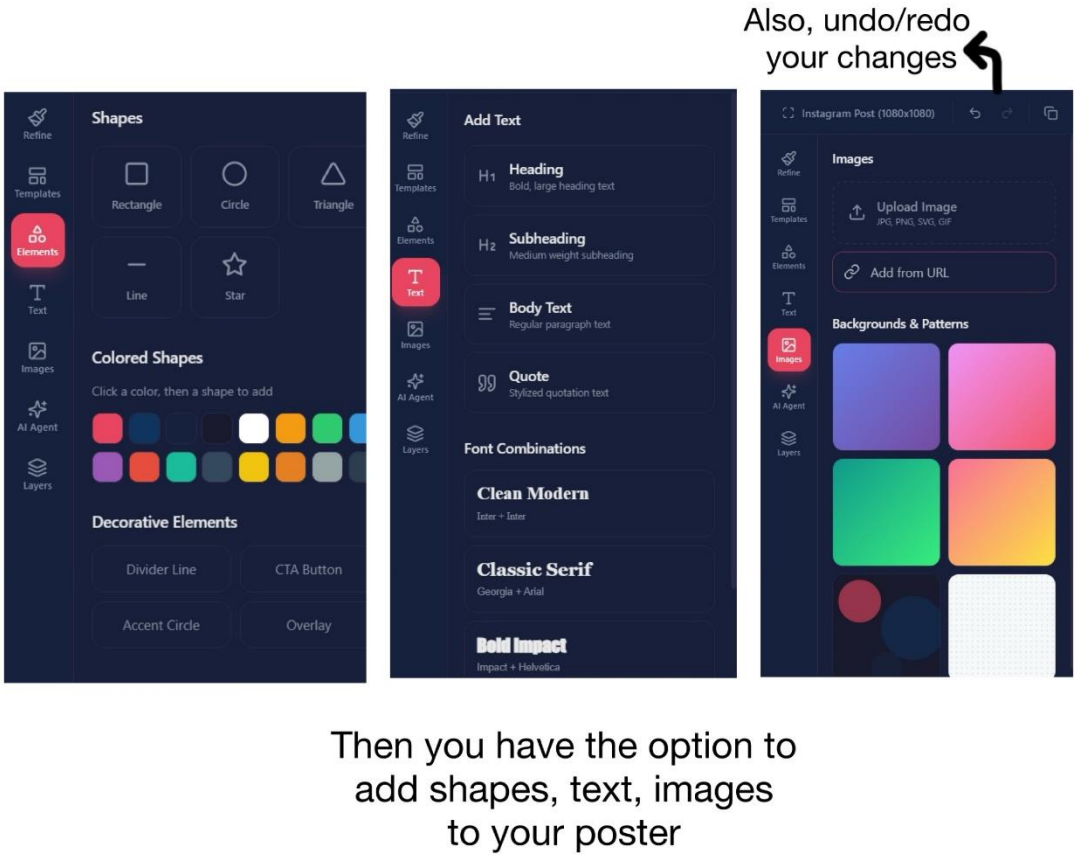
We can search for other background image, or can upload one from our system-



Other customisation options include- adding a template for offers and sales with pre-defined layouts, which can be edited and deleted by the user-



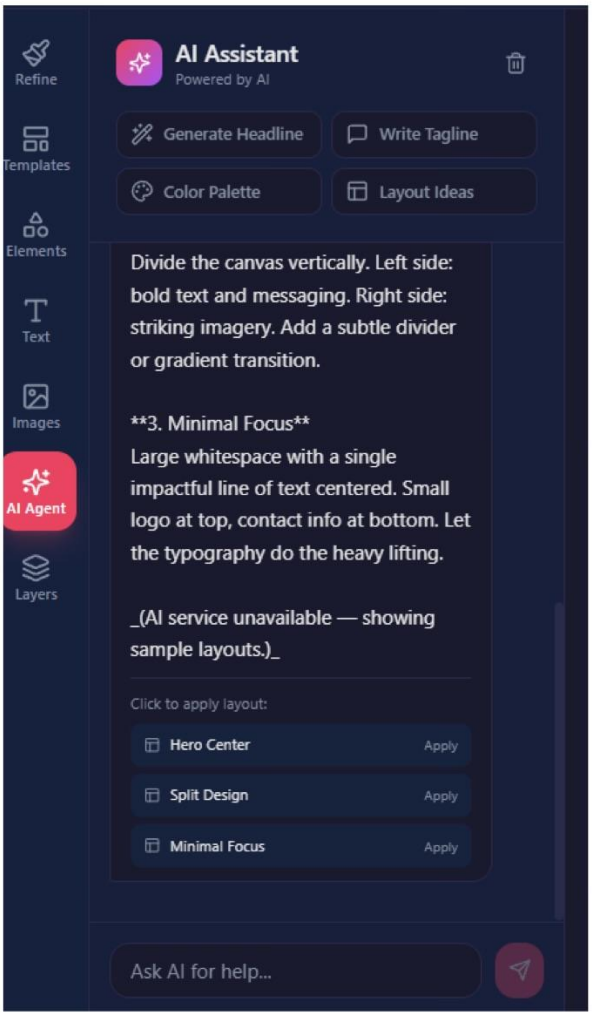
User can also add shapes, text and images (can choose among provided images, can upload any, can use an image URL as well). Each of these applied layers can be edited/deleted by the options on the right panel as shown in previous screenshot. User has the option to undo/redo the changes as well.



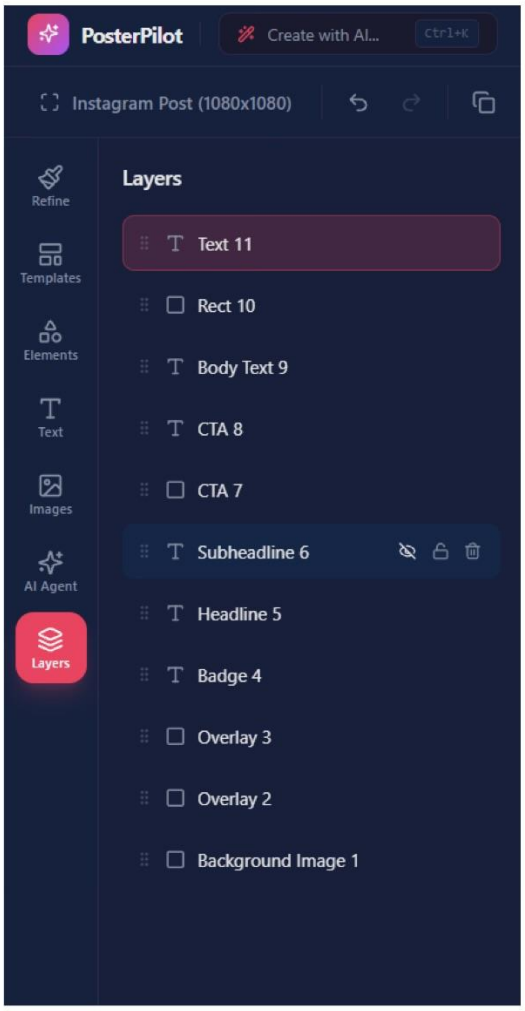
Then you have the option to add shapes, text, images to your poster

Now comes the AI Agent help option to add headlines, taglines, change layout and color palettes to the poster with a single click.

After all the changes, user can navigate to 'Layers' tab to hide/unhide any element of the poster (useful to hide past offer/discount), lock any element for further changes (like the brand logo), and delete an element.

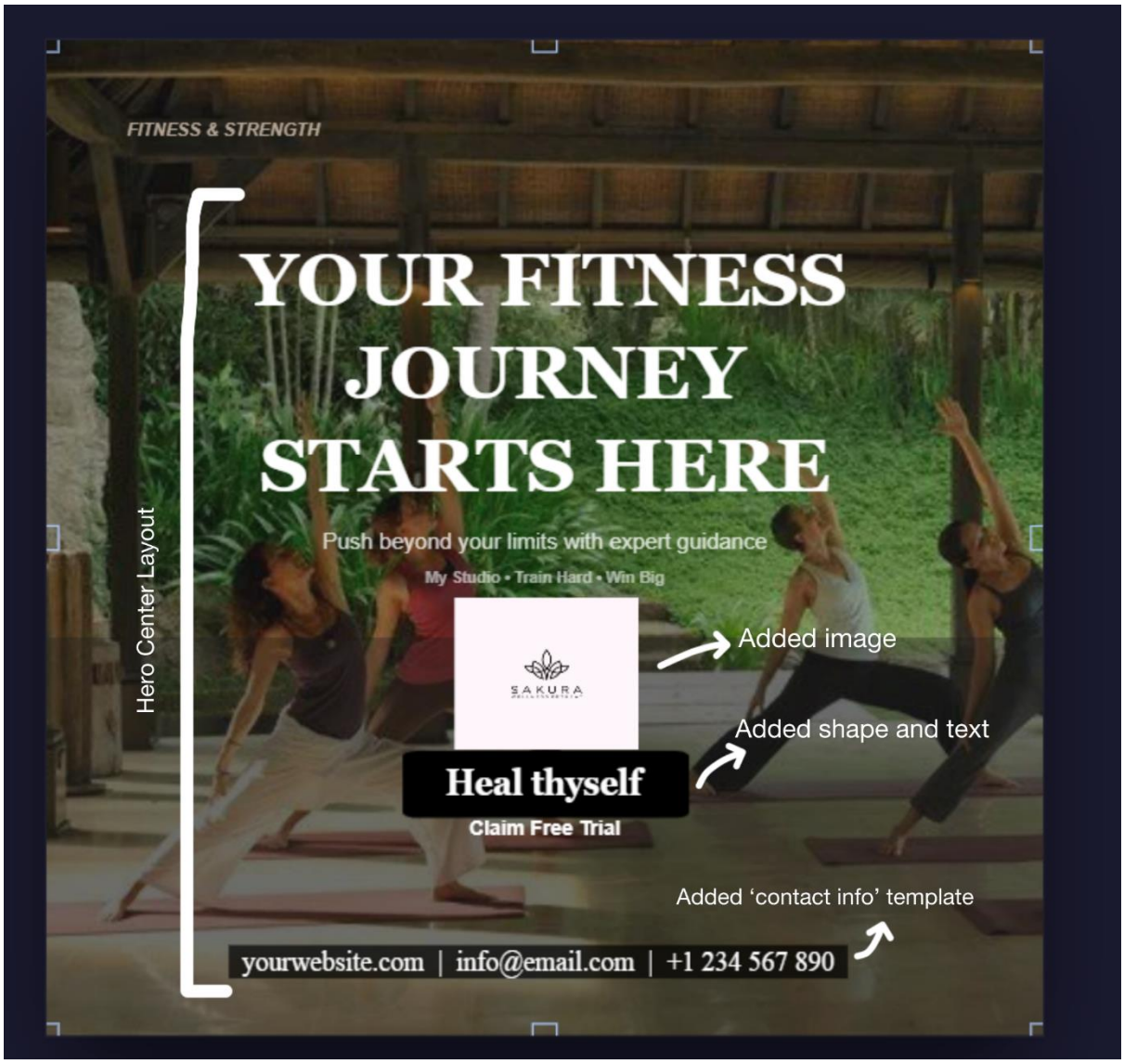
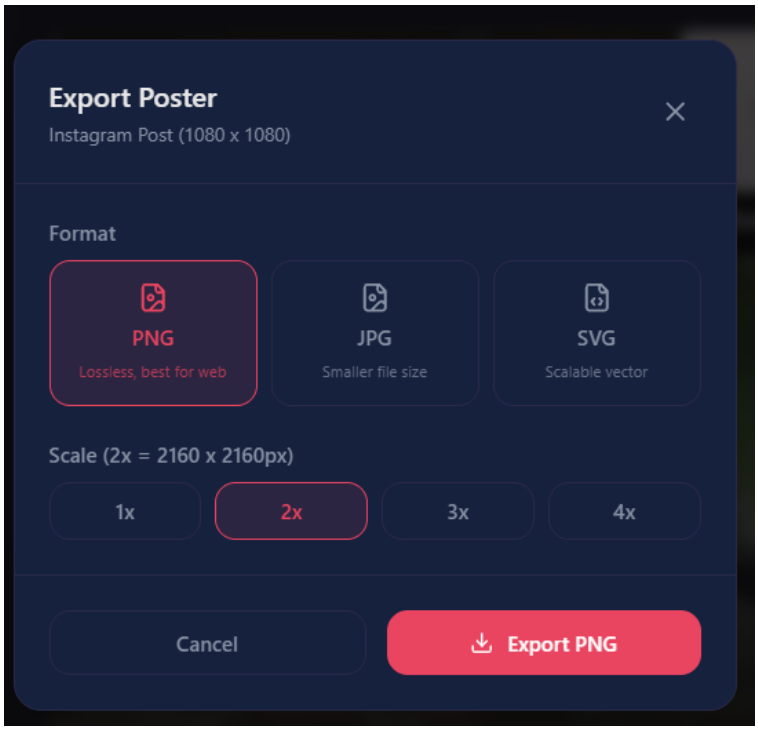


Can apply headline, tagline, layout suggestions by AI through one click



Can hide/unhide, lock (to avoid any further edits or deletion), delete any element of the poster

Here’s an example of the final poster with the above customisation options used.



User can export the poster in different formats available on the top right corner of the screen to open the below dialog box.